

PETHANARAJ S

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Professional Summary

Mobile Application Developer with 3+ years of experience building and delivering high-performance, enterprise-grade mobile applications using **Flutter** and **Android Native (Kotlin)**. Proven expertise in cross-platform and native mobile development, Clean Architecture, MVVM, state management (BLoC · Riverpod · GetX), Firebase integration, and RESTful API consumption. Skilled in crafting pixel-perfect UIs, optimising app performance, implementing offline-first strategies, and managing end-to-end production releases on the **Google Play Store** and **Apple App Store**. Adept at delivering scalable enterprise mobile solutions with a strong focus on user experience, code maintainability, and Agile delivery.

Work Experience

Hinduja Tech Limited April 2025 – Present

Role – Mobile Application Developer (Flutter | Android | Kotlin) | Client: Ashok Leyland

- ✓ Developing cross-platform Flutter mobile applications for enterprise fleet and asset management workflows
- ✓ Building native Android screens using **Kotlin** and **Jetpack Compose** for performance-critical modules
- ✓ Implementing **Clean Architecture** with MVVM pattern ensuring testable, maintainable, and scalable codebases
- ✓ Integrating **RESTful APIs** with secure JWT-based authentication and role-based access flows
- ✓ Managing local data persistence using **Room Database** and **Hive** for offline-first functionality
- ✓ Handling real-time GPS tracking and map-based UI features within the Flutter mobile application
- ✓ Conducting mobile performance profiling and optimisation to ensure smooth 60fps UI rendering
- ✓ Managing app releases and hotfixes through **Firebase App Distribution**, Play Store, and App Store pipelines

Solvedge Inc Dec 2022 – March 2025

Role – Mobile Application Developer (Flutter | Android | Kotlin)

- ✓ Designed and delivered 6+ production mobile applications across **Android**, **iOS**, and **Flutter Web**
- ✓ Built cross-platform UIs using Flutter with **BLoC**, **Riverpod**, and **GetX** state management
- ✓ Developed native Android features using **Kotlin**, **Android SDK**, and **Jetpack** components
- ✓ Integrated **Firebase Cloud Messaging (FCM)** for real-time push notifications across multiple apps
- ✓ Implemented offline storage solutions using **Room Database**, **Hive**, and **SharedPreferences**
- ✓ Consumed and integrated **RESTful APIs** with Dio/Retrofit; handled authentication and token refresh flows
- ✓ Built reusable widget libraries and component-driven UI following Material Design and custom design systems
- ✓ Implemented deep-linking, QR-code scanning, camera, GPS, and biometric features using platform channels
- ✓ Managed CI/CD pipelines using **Git**, AWS CodeCommit, and Firebase App Distribution for automated releases
- ✓ Collaborated in Agile/Scrum teams; participated in sprint planning, code reviews, and UAT cycles

Core Technical Expertise

✓ Mobile Technologies:	Flutter · Android SDK · Jetpack Compose · Cross-Platform App Development · Android Native
✓ Android Development:	Kotlin · Android SDK · Jetpack (ViewModel · LiveData · Navigation · WorkManager) · Coroutines
✓ Flutter Ecosystem:	Dart · Flutter Widgets · Custom UI · Platform Channels · Flutter Web · Flutter Animations
✓ State Management:	BLoC · Riverpod · GetX · Provider · ChangeNotifier
✓ Architecture & Patterns:	Clean Architecture · MVVM · Repository Pattern · Dependency Injection
✓ Firebase & Notifications:	Firebase Cloud Messaging · Firebase Firestore · Firebase Auth · Firebase App Distribution
✓ API Integration:	RESTful APIs · Dio · Retrofit · JWT Authentication · OAuth2 · Role-Based Access Control
✓ Local Storage & DB:	Room Database · Hive · SharedPreferences · SQLite
✓ Tools & Deployment:	Android Studio · VS Code · Xcode · Git · CI/CD · Play Store · App Store · Postman · Agile/Scrum

Key Projects

✓ Tool Tracking Mobile App | Flutter · Kotlin · Android SDK · Room · GPS · REST APIs · Firebase · MVVM

Enterprise asset lifecycle management app for field engineers and vendors at Ashok Leyland.

- ✓ Developed the Flutter mobile client for field engineers with real-time GPS-based tool location tracking
- ✓ Built Flutter Web admin dashboard with role-based views for managers, vendors, and administrators
- ✓ Implemented image-based audit verification workflows with camera integration and cloud upload
- ✓ Designed offline-capable data sync using **Room Database** and background sync with WorkManager

- ✓ Delivered 15+ API integration points with JWT-secured endpoints; handled token refresh and error states
- ✓ Enabled **Clean Architecture** with BLoC state management for scalable, testable feature modules
- ✓ Achieved smooth 60fps UI performance through lazy loading, widget caching, and render optimisations

✓ **Visitor Management Mobile App** | Flutter · Kotlin · JWT · QR Scanner · REST APIs · MVVM · Clean Architecture

Full-featured visitor access control mobile application for high-security manufacturing facilities.

- ✓ Built Flutter mobile app for visitors, security staff, admins, and approvers with distinct role-based UIs
- ✓ Integrated **QR-code scanning** for fast visitor check-in, reducing processing time by **60%**
- ✓ Implemented multi-step approval workflow with real-time status updates using push notifications
- ✓ Developed JWT-based secure authentication with biometric login support on Android
- ✓ Used **GetX** for state management and navigation, enabling rapid development of approval flows
- ✓ Delivered production releases on both **Android (Play Store)** and **iOS (App Store)**

✓ **IAM Medical Guidelines App** | Flutter · Firebase FCM · Push Notifications · Cross-Platform · Play Store · App Store

Production-grade clinical reference mobile application for healthcare professionals in Canada.

- ✓ Developed a polished cross-platform Flutter app with rich content rendering for clinical guidelines
- ✓ Integrated **Firebase Cloud Messaging (FCM)** for real-time push notifications on guideline updates
- ✓ Implemented secure content delivery with offline caching using Hive for uninterrupted access
- ✓ Managed full **Play Store** and **App Store** release lifecycle including metadata, versioning, and phased rollouts
- ✓ Optimised app startup time and content load performance for low-bandwidth healthcare environments

✓ **AI Nurse Assistant Chatbot App** | Flutter · Speech-to-Text · Text-to-Speech · Conversational AI · Mobile UI

AI-powered healthcare voice assistant mobile app enabling natural patient–system interaction.

- ✓ Built an intuitive chat-based Flutter UI with real-time voice interaction and animated message flows
- ✓ Integrated **Speech-to-Text** and **Text-to-Speech** for bidirectional voice-based patient query handling
- ✓ Designed conversational state flow with BLoC ensuring responsive, context-aware interactions
- ✓ Implemented accessibility-focused UI patterns for diverse patient demographics

Education

- ✓ B.E. Mechanical Engineering, Anna University (2018 – 2022) – 82%
- ✓ Higher Secondary, AVM Hr. Sec. School (2016 – 2018) – 84%
- ✓ SSLC, Govt High School – 90%

Achievements

- ✓ Successfully delivered and maintained 6+ production mobile applications across Android, iOS, and Flutter Web, managing releases, updates, and hotfixes through Play Store, App Store, and Firebase App Distribution.
- ✓ **Best Debut of the Year** – Awarded for exceptional performance and rapid impact in delivering high-quality mobile applications during the first year at the organisation.
- ✓ Delivered 6+ production applications across Android, iOS, and Flutter Web serving enterprise clients.